

OriGround: Orientation-Aware Neuro-Symbolic Zero-Shot 3D Visual Grounding

Anonymous ACL submission

Abstract

Zero-shot 3D visual grounding (3DVG) aims to localize a referred object in a 3D scene without task-specific grounding supervision. Existing training-free methods commonly reason over object categories, centers, and bounding boxes, but these representations do not encode the local reference frames needed to interpret viewpoint-dependent expressions such as left of, behind, or facing the door. We propose OriGround, an orientation-aware neuro-symbolic framework that explicitly models object-centric orientations and infers viewpoints into zero-shot 3DVG. OriGround parses each query into a viewpoint-aware symbolic program, estimates object-centric orientation bases from multi-view observations, performs anchor- and observer-specific geometric relation scoring, and constructs a reference-frame-aligned visual prompt for final VLM-based disambiguation. On Nr3D, OriGround improves over the strongest training-free baseline by 8.4 points in overall accuracy, with larger gains of 8.9 points on hard queries and 5.6 points on view-dependent queries. On ScanRefer, OriGround further improves overall Acc@0.5 by 10 points.

1 Introduction

Zero-shot 3D visual grounding (3DVG) aims to localize a target object in a 3D scene according to a natural-language instruction, without task-specific training on 3D grounding annotations (Achlioptas et al., 2020; Yuan et al., 2024b). While many referring expressions can be resolved from object categories, attributes, and metric distances, viewpoint-dependent spatial language remains particularly challenging. For example, the phrase “the chair on the left of the sofa” may refer to the sofa’s intrinsic left side, whereas “the chair on the left when facing the sofa” depends on the observer’s viewpoint. In both cases, the word “left” cannot be interpreted solely in a fixed world coordinate system. Instead,

it requires an explicit reference frame defined by either an anchor object, an observer pose, or both.

Recent zero-shot 3DVG methods have made substantial progress by leveraging large language models (LLMs) and vision-language models (VLMs) for language decomposition, symbolic program synthesis, and visual-semantic reasoning. Since pretrained LLMs and VLMs are not directly designed to operate on raw 3D scenes with metric structure, these methods usually convert a scene into object-centric representations, such as semantic categories, instance identifiers, 3D bounding boxes, and center coordinates (Fang et al., 2024; Yuan et al., 2024b; Mi et al., 2025). Such representations are effective for category matching and many view-independent relations. However, they do not encode the local directional bases needed to distinguish front, back, left, and right with respect to a specific object or observer.

To mitigate viewpoint-related ambiguity, recent VLM-based methods further incorporate 2D views or selected viewpoints into the grounding process (Li et al., 2025; Jin et al., 2025). However, rendered views alone do not explicitly specify the reference frame under which a directional relation should be evaluated. The same scene may support different interpretations depending on whether the relation is world-centric, anchor-centric, or observer-centric. Consequently, viewpoint reasoning may still rely on the assumptions made by the VLM from language, rendering choices, or dataset biases. This limitation becomes especially severe for expressions involving left, right, front, behind, facing the door, or with your back to the window.

To address these limitations, we propose **OriGround**, an orientation-aware neuro-symbolic framework for zero-shot 3DVG. The key idea is to make the reference frame explicit throughout both symbolic reasoning and visual prompting. OriGround first parses the natural-language instruction into a viewpoint-aware symbolic program that

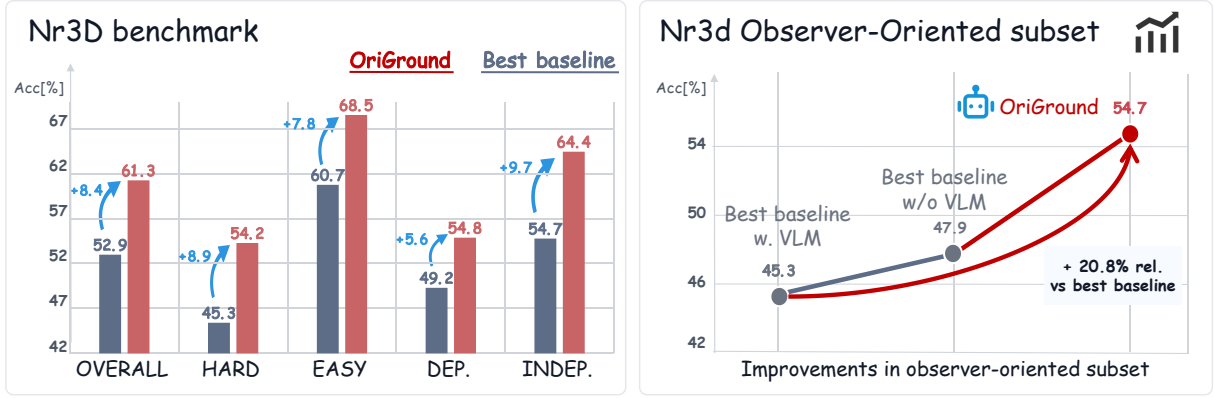


Figure 1: Performance overview of OriGround on Nr3D. Left: comparison with the previous best training-free baseline on standard Nr3D splits. Right: results on the Observer-Oriented subset with explicit observer cues, where OriGround reaches 54.7% accuracy and obtains a 20.8% relative gain.

identifies the target category, spatial relations, anchor objects, and observer cues. It then estimates object-centric orientation bases from multi-view object observations and aligns them with the world coordinate system. These orientations define intrinsic up, left, and front directions for each object instance, allowing directional predicates to be evaluated in anchor-specific or observer-specific reference frames. Finally, OriGround constructs a reference-frame-aligned visual prompt that combines multi-view candidate evidence with a top-down map rotated to the inferred observer perspective, enabling the VLM to perform final disambiguation under the same reference frame used by the symbolic executor.

Experiments on Nr3D and ScanRefer demonstrate that OriGround improves over strong training-free baselines. As summarized in Fig. 1, OriGround achieves an 8.4-point overall gain over the previous best training-free baseline on Nr3D, with particularly large improvements on hard, view-dependent, and observer-oriented queries. These results suggest that explicit orientation modeling is not merely an auxiliary visual cue but a useful geometric prior for defining the reference frames under which spatial language should be interpreted.

In summary, the main contributions of this work are as follows:

- We formulate viewpoint-dependent zero-shot 3D visual grounding as a reference-frame ambiguity problem, where bounding-box-only scene representations are insufficient for interpreting directional relations such as left, right, front, and behind.
- We propose an orientation-aware neuro-symbolic reasoning framework that estimates object-centric 3D orientation bases

and uses them to score spatial relations in anchor-specific and observer-specific reference frames.

- We introduce reference-frame-aligned visual prompting, which combines multi-view candidate evidence with a dynamically rotated top-down map, and demonstrate consistent improvements over strong training-free baselines on Nr3D and ScanRefer.

2 Related Work

Zero-shot 3D Visual Grounding. Zero-shot 3DVG aims to localize referred objects in 3D scenes without task-specific supervision, reducing reliance on costly 3D grounding annotations (Yuan et al., 2024b; Yang et al., 2024). Existing training-free methods mainly leverage pretrained LLMs or VLMs. LLM-based methods convert 3D scenes into textual or symbolic representations and perform tool-augmented spatial reasoning (Yang et al., 2024; Yuan et al., 2024b,a; Fang et al., 2024), but they lack fine-grained visual evidence and ignore intrinsic object orientations. VLM-based methods introduce rendered views, scene images, or visual prompts for visual-semantic grounding (Li et al., 2025; Xu et al., 2025; Mi et al., 2025; Jin et al., 2025); however, they still rely largely on VLMs to infer 3D viewpoints, which often leads to errors in viewpoint-dependent relations. In contrast, OriGround explicitly extracts object-centric 3D orientations, uses them for deterministic neuro-symbolic reasoning, and aligns visual prompts with the inferred observer perspective.

Neuro-Symbolic Reasoning. Neuro-symbolic reasoning integrates neural perception with the compositionality and interpretability of symbolic reasoning (Andreas et al., 2016; Mao et al., 2019;

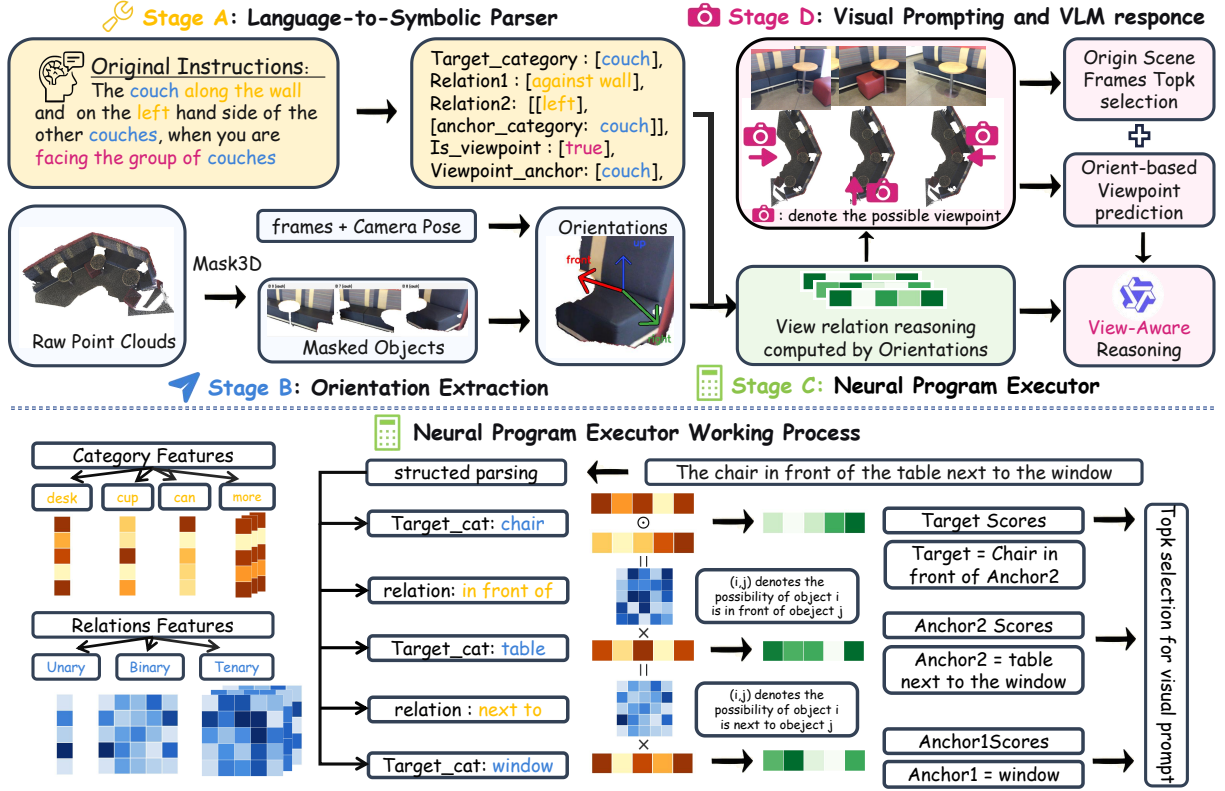


Figure 2: Overview of OriGround. Stage A parses the natural-language instruction into a viewpoint-aware symbolic program. Stage B extracts object-centric orientations from masked multi-view observations. Stage C executes the neural-symbolic program to score relations and filter candidates. Stage D builds a viewpoint-aware visual prompt and lets the VLM make the final grounding decision.

Yi et al., 2018). Representative methods, such as the Neuro-Symbolic Concept Learner, ground symbolic programs in learned visual concepts, achieving strong data efficiency and compositional generalization (Mao et al., 2019). Recent works further extend this paradigm to 3D scene understanding and grounding, where NS3D performs compositional reasoning over 3D scene-graph representations (Hsu et al., 2023; Feng et al., 2024), and Transcrib3D and LaSP employ LLMs to parse language into executable spatial programs (Fang et al., 2024; Mi et al., 2025). However, these methods mainly rely on positions and bounding-box relations, leaving object-centric orientations unmodeled. OriGround instead injects intrinsic 3D orientations into neuro-symbolic reasoning for reference-frame-aware relation scoring.

3 Methodology

3.1 Problem Formulation and Overview

Problem Formulation. Zero-shot 3D visual grounding aims to localize the target object in a 3D scene from a natural-language query, without task-specific training. Given a scene $\mathcal{S}$ and a query $\mathcal{Q}$, the model selects the referred target $\hat{T}$ from

candidate objects $\mathcal{O} = \{O_i\}_{i=1}^N$. Each candidate O_i is represented by its 3D point mask, semantic category, bounding box B_i , and center coordinate $\mathbf{c}_i$. For Nr3D, $\hat{T}$ is evaluated as an object identifier; for ScanRefer, it is evaluated by the corresponding 3D bounding box. Since bounding boxes lack intrinsic object directions, they are insufficient for viewpoint-dependent expressions such as *left of*, *behind*, and *facing*. OriGround augments each object with an orientation basis $\mathbf{R}_i = [\mathbf{v}_u^{(i)}, \mathbf{v}_l^{(i)}, \mathbf{v}_f^{(i)}]$, denoting its intrinsic up, left, and front directions. Based on this, OriGround performs viewpoint-aware parsing, orientation-aware symbolic reasoning, and viewpoint-aligned visual prompting to select the target.

Overall Pipeline. To address the ambiguity from missing object-centric directions, we propose OriGround, an orientation-aware neuro-symbolic framework for zero-shot 3D visual grounding. OriGround introduces object-centric orientations to model view-dependent spatial relations and observer cues. As illustrated in Fig. 2, OriGround follows a four-stage inference flow. First, the Language-to-Symbolic Parser (Sec. 3.2) converts the query $\mathcal{Q}$ into a viewpoint-aware symbolic pro-

gram $\mathcal{P}$. Second, Object-Centric Orientation Estimation (Sec. 3.3) extracts an orientation basis $\mathbf{R}_i$ for each candidate object O_i . Third, Orientation-Aware Symbolic Reasoning (Sec. 3.4) evaluates spatial relations under the corresponding reference frames and produces geometry-based grounding scores. Finally, Holistic Visual Prompting with Viewpoint Alignment (Sec. 3.5) constructs multi-view candidate evidence and a viewpoint-aligned top-down map for the final VLM decision.

3.2 Language-to-Symbolic Parser

Grounding natural-language queries in 3D scenes is challenging due to relational dependencies and implicit references. To bridge this semantic-to-geometric gap, our neuro-symbolic parser uses an LLM to translate the unstructured query Q into an executable, structured symbolic program $\mathcal{P}$. This intermediate representation explicitly decouples reasoning steps by encapsulating object categories and their spatial constraints.

Specifically, the symbolic program $\mathcal{P}$ contains three core fields: category, relations, and viewpoint. The category field identifies target or anchor objects, the relations field stores spatial predicates and their anchors, and the viewpoint field indicates whether Q contains a subjective perspective and, if so, its viewpoint anchor.

Due to the unconstrained nature of human language, raw spatial descriptions often form a long-tail distribution. To alleviate this linguistic ambiguity, we prompt the LLM to map the original terms in Q into a predefined vocabulary of standardized relation labels for $\mathcal{P}$. To disambiguate viewpoint-dependent queries (e.g., “facing from the foot of the bed”), the parser explicitly extracts the viewpoint component. This mechanism systematically identifies the presence of an egocentric perspective and isolates its corresponding reference anchor. Ultimately, transforming the unstructured input into this viewpoint-aware hierarchical program empowers the downstream modules to compute geometric scores, execute spatial reasoning, and explicitly predict the observer’s viewpoint for the VLM.

3.3 Object-Centric Orientation Estimation Orientation Inference and Global Alignment.

Orientations, sizes, and positions inherently determine spatial relations. For each object O_i , we isolate its visual appearance in multi-view images using object-level masks projected from the 3D instance mask. The masked image $\hat{I}_i^{(j)}$ is processed

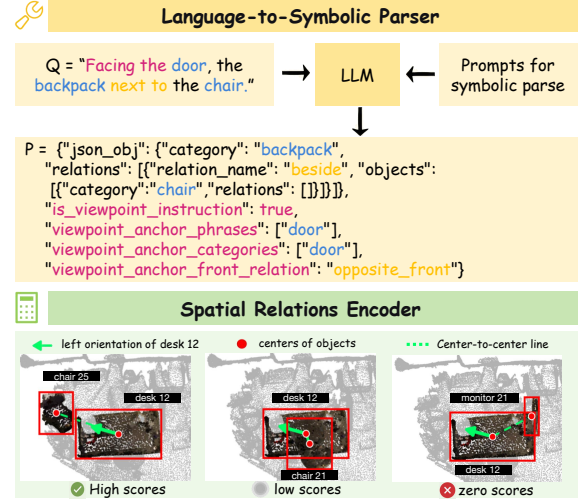


Figure 3: Language-to-symbolic parsing and spatial relation encoding. Top: the parser converts a natural-language query into a structured symbolic program. Bottom: the encoder computes relation scores from object coordinates and intrinsic anchor orientations.

by Orient-Anything-V2 (Wang et al., 2025), which predicts a local rotation matrix $R_{\text{local}} \in \text{SO}(3)$ representing the object’s orientation relative to the camera’s perspective. When multiple views are available, we use the view with the largest visible object area for orientation inference.

We extract the canonical front vector $n_f \in \mathbb{R}^3$ from R_{local} and transform it into the world coordinate system using camera extrinsic rotation $R_{\text{ext}}^{(j)}$:

$$\mathbf{v}_{f,\text{raw}}^{(i)} = R_{\text{ext}}^{(j)} \cdot n_f^{(i)} \quad (1)$$

Considering that indoor objects are typically gravity-aligned, we fix the up vector as $\mathbf{v}_u^{(i)} = [0, 0, 1]^\top$. To eliminate estimation noise in the pitch axis, we project $\mathbf{v}_{f,\text{raw}}^{(i)}$ onto the horizontal plane to obtain the orthogonal front and left vectors:

$$\begin{aligned} \tilde{\mathbf{v}}_f^{(i)} &= \mathbf{v}_{f,\text{raw}}^{(i)} - \left(\mathbf{v}_{f,\text{raw}}^{(i)\top} \mathbf{v}_u^{(i)} \right) \mathbf{v}_u^{(i)}, \\ \mathbf{v}_f^{(i)} &= \frac{\tilde{\mathbf{v}}_f^{(i)}}{\|\tilde{\mathbf{v}}_f^{(i)}\|_2}, \quad \mathbf{v}_l^{(i)} = \mathbf{v}_u^{(i)} \times \mathbf{v}_f^{(i)}. \end{aligned} \quad (2)$$

The resulting object-centric orientations $\mathbf{R}_i = [\mathbf{v}_u^{(i)}, \mathbf{v}_l^{(i)}, \mathbf{v}_f^{(i)}]$ can then be utilized to compute view-dependent geometric relation scores.

3.4 Orientation-Aware Symbolic Reasoning Spatial Relations Encoder. For each directional predicate, the executor first determines its reference frame from the symbolic program: world-centric predicates are evaluated in the global frame, anchor-centric predicates in the anchor object’s local frame,

Algorithm 1 Recursive Executor

Require: Symbolic node $\mathcal{N}$, category features C , relation features R

Output: Grounding score s

```

1:  $c \leftarrow \mathcal{N}$ ["category"]
2:  $s \leftarrow C[c]$ 
3: for each  $rel \in \mathcal{N}$ ["relations"] do
4:    $r \leftarrow rel$ ["name"]
5:    $a \leftarrow rel$ ["anchor"]
6:   if  $a$  is a nested symbolic node then
7:      $s_a \leftarrow \text{Execute}(a, C, R)$  {Recursive top-down descent}
8:   else
9:      $s_a \leftarrow C[a]$  {Base leaf anchor}
10:  end if
11:   $M_r \leftarrow R[r]$ 
12:   $s \leftarrow s \odot (M_r \times s_a)$  {Bottom-up score propagation}
13: end for
14: Return  $s$ 

```

and observer-centric predicates in the inferred observer frame. The encoder is implemented as executable functions using extracted orientations to compute relational geometric scores. For view-independent relations like *large*, *near*, or *above*, the geometric scores can be directly derived using bounding boxes and center coordinates of the objects. However, these absolute measurements are insufficient for view-dependent relations such as *left* or *behind*. Such relations strictly require the explicit orientation information extracted in our previous module.

Taking the “left” relation as an example (as conceptually illustrated in Fig. 3), let $\mathbf{c}_i, \mathbf{c}_j \in \mathbb{R}^3$ denote the spatial centers of candidate O_i and anchor O_j , respectively. Utilizing the anchor’s local left-vector $\mathbf{v}_l^{(j)}$ extracted from its orientation $\mathbf{R}_j$, the normalized direction vector $\mathbf{v}_{ji}$ from the anchor to the candidate, and the final geometric score S_{left} are computed as follows:

$$\mathbf{v}_{ji} = \frac{\mathbf{c}_i - \mathbf{c}_j}{\|\mathbf{c}_i - \mathbf{c}_j\|_2}, \quad (3)$$

$$S_{\text{left}}(i, j) = \text{ReLU}(\mathbf{v}_{ji} \cdot \mathbf{v}_l^{(j)}).$$

This formulation evaluates the anchor-centric *left* relation in the local coordinate frame of the anchor object, ensuring that spatial reasoning is consistent with the anchor orientation. For observer-centric directional relations, we replace the anchor-local direction vector with the corresponding direction vector from the inferred observer frame; for instance, $\mathbf{v}_l^{(j)}$ is replaced by the observer-left direction $\mathbf{v}_l^{\text{obs}}$. **Neural Program Executor.** While the parser decomposes queries top-down from the primary target to leaf anchors, geometric scoring propagates

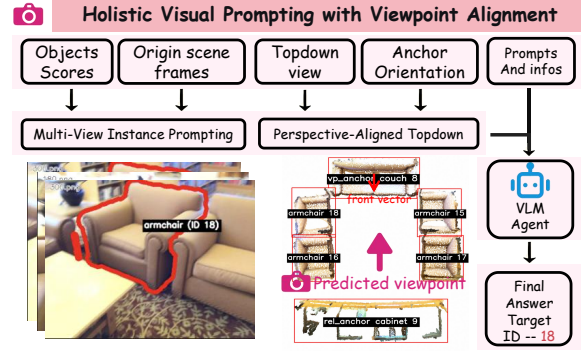


Figure 4: Holistic visual prompting with viewpoint alignment. The VLM receives candidate instance evidence together with a perspective-aligned top-down map to synthesize a final response.

strictly bottom-up. Starting from the leaf nodes, each anchor’s grounding score s_a is multiplied by its spatial relation matrix M_r , which is computed by the Relation Encoder. This contextual score is then element-wise multiplied (denoted as $\odot$) by the target’s initial category score s . By propagating these scores bottom-up, we step-by-step verify all spatial constraints and filter out mismatched candidates and anchors, as detailed in Algorithm 1.

Leveraging the final computed scores, we isolate the top- K candidates along with their high-confidence anchors, paving the way for the subsequent VLM prompt construction and viewpoint prediction.

3.5 Holistic Visual Prompting with Viewpoint Alignment

To leverage the VLM’s proficiency in fine-grained feature discrimination and to reduce visual ambiguities through viewpoint prediction, we construct a hybrid prompt comprising instance-level scene images augmented with a viewpoint-aligned global context as illustrated in Fig. 4.

Multi-View Instance Prompting (MIP). After pruning the search space to the top- K candidates, we build a compact 2×2 visual prompt for each candidate. One view is the frame where the candidate occupies the largest visible area, while the other is chosen to maximize co-visibility with the relevant anchor whenever possible. The top row keeps the original frames, and the bottom row shows the corresponding annotated views, giving the VLM both clear candidate appearance and explicit local relational evidence.

Perspective-Aligned Global Context (PGC). To provide a holistic spatial layout, we render a top-down map $\mathcal{M}_{\text{td}}$ using only the shortlisted candidates and the relation-relevant anchors retained by

the symbolic executor. This reduces irrelevant clutter while preserving the geometry required by the query. We then rotate the map to the observer perspective θ_{obs} , which is derived from the symbolic program $\mathcal{P}$ and the orientation of the reference anchor. The resulting perspective-aligned map is denoted as $\widehat{\mathcal{M}}_{\text{td}}$.

Orientation-Hint Rendering. In addition to candidate positions, we render the front directions of relation-relevant anchors on the perspective-aligned top-down map. These orientation hints expose the local reference frames used by the symbolic executor and help the VLM interpret directional relations such as *left*, *right*, and *behind*. We denote the final annotated map as $\widehat{\mathcal{M}}_{\text{td}}$.

Final VLM Decision. The textual query $\mathcal{Q}$, candidate-specific instance prompt $\mathcal{V}_{\text{inst}}^{(i)}$, and annotated top-down map $\widehat{\mathcal{M}}_{\text{td}}$ are jointly fed into the VLM. The final target is selected from the top- K candidates:

$$\hat{T} = \underset{O_i \in \mathcal{O}_{\text{top-}K}}{\operatorname{argmax}} P_{\text{VLM}}(O_i \mid \mathcal{Q}, \mathcal{V}_{\text{inst}}^{(i)}, \widehat{\mathcal{M}}_{\text{td}}). \quad (4)$$

These inputs allow the VLM to compare shortlisted candidates using both local appearance evidence and global spatial layout.

4 Experiments

4.1 Experimental Settings

Datasets. We evaluate OriGround on two standard 3D visual grounding benchmarks: Nr3D from ReferIt3D (Achlioptas et al., 2020) and ScanRefer (Chen et al., 2020). Nr3D contains 41,503 human-written descriptions and is reported on the *Easy*, *Hard*, *View-dependent*, and *View-independent* subsets, where *Hard* indicates stronger same-category ambiguity and *View-dependent* indicates observer-centric spatial language. Since Nr3D provides ground-truth 3D bounding boxes, evaluation amounts to identifying the correct target object ID. ScanRefer contains 51,583 expressions from 800 ScanNet scenes and reports Acc@0.25 and Acc@0.5 on the *Overall*, *Unique*, and *Multiple* splits. Unlike Nr3D, ScanRefer requires predicting a target 3D bounding box and evaluates grounding accuracy by IoU.

Implementation Details. In all experiments, the language-to-symbolic parser and the visual grounding stage use Qwen-VL-Max (Bai et al., 2023). Object orientations are estimated with Orient-Anything-V2 (Wang et al., 2025) and projected

into the world coordinate system as described in Sec. 3.4. For ScanRefer, we follow VLM-Grounder (Xu et al., 2025) and evaluate on the same subset. For the analysis in Table 4, we construct the Observer-Oriented subset of Nr3D (Nr3D-VP) using our symbolic parser. It contains 1,859 queries whose symbolic programs include explicit observer cues or viewpoint anchors.

4.2 Comparative Study

Nr3D. Table 1 reports the results on Nr3D. Among training-free methods, OriGround achieves the best performance across all standard splits, improving over LaSP (Mi et al., 2025) by 8.4 points in overall accuracy. The gains are especially clear on the Hard and View-dependent subsets, indicating that explicit orientation modeling and viewpoint-aligned prompting are effective for same-category ambiguity and observer-dependent spatial language. Notably, OriGround also approaches strong supervised methods and surpasses some established supervised baselines, despite using no task-specific grounding supervision.

ScanRefer. Table 2 evaluates OriGround on ScanRefer using the same 250-sample subset as VLM-Grounder (Xu et al., 2025). OriGround obtains the best overall training-free performance, with a particularly large gain on Acc@0.5. Compared with Nr3D, the improvement on ScanRefer is more moderate, mainly because ScanRefer is evaluated using predicted 3D proposals rather than ground-truth object boxes. Detector localization errors and missing proposals therefore bound the final accuracy.

4.3 Ablation Study

Ablation of Architecture Design. Table 3 summarizes the contribution of the main visual-prompting components in Sec. 3.5: Multi-View Instance Prompting (MIP), Perspective-Aligned Global Context (PGC), and the final Orientation-Hint rendered on the top-down map.

- **(a) Symbolic Filtering.** Without the final VLM stage, symbolic filtering alone reaches 50.0% overall accuracy. This shows that symbolic reasoning already removes many inappropriate candidates, but it is not enough for fine-grained disambiguation.
- **(b) Instance-Centric Prompting.** Adding the VLM together with MIP raises accuracy from 50.0% to 58.6%. This contributes the largest gain in the table, showing that candidate visual evidence is crucial after symbolic

Table 1: Comparison of 3DVG results on Nr3D. The Supervised column denotes whether each method uses task-specific supervised training.

Method	Supervised	Overall	Easy	Hard	View Dep.	View Indep.
ViL3DRef (Chen et al., 2022)	✓	64.4	70.2	57.4	62.0	64.5
CoT3DRef (Bakr et al., 2024)	✓	64.4	70.0	59.2	61.9	65.7
3D-VisTA (Zhu et al., 2023)	✓	64.2	72.1	56.7	61.5	65.1
BUTD-DETR (Jain et al., 2022)	✓	54.6	60.7	48.4	46.0	58.0
SAT (Yang et al., 2021)	✓	49.2	56.3	42.4	46.9	50.4
ZSVG3D* (Yuan et al., 2024b)	×	40.2	49.1	31.1	37.8	41.6
SeeGround (Li et al., 2025)	×	46.1	54.5	38.3	42.3	48.2
VLM-Grounder (Xu et al., 2025)	×	48.0	55.2	39.5	45.8	49.4
LaSP w/o VLM (Mi et al., 2025)	×	50.7	58.7	43.0	45.6	53.2
LaSP (Mi et al., 2025)	×	52.9	60.7	45.3	49.2	54.7
OriGround (ours)	×	61.3	68.5	54.2	54.8	64.4

Table 2: Comparison of 3DVG results on ScanRefer. The Supervised column denotes whether each method uses task-specific supervised training.

Method	Supervised	Overall		Unique		Multiple	
		Acc@0.25	Acc@0.5	Acc@0.25	Acc@0.5	Acc@0.25	Acc@0.5
ScanRefer (Chen et al., 2020)	✓	37.3	24.3	65.0	43.3	30.6	19.8
TGNN (Huang et al., 2021)	✓	34.3	29.7	64.5	53.0	27.0	21.9
InstanceRefer (Yuan et al., 2021)	✓	40.2	32.9	77.5	66.8	31.3	24.8
3DVG-Transformer (Zhao et al., 2021)	✓	47.6	34.7	81.9	60.6	39.3	28.4
BUTD-DETR (Jain et al., 2022)	✓	52.2	39.8	84.2	66.3	46.6	35.1
OpenScene (Peng et al., 2023)	×	13.2	6.5	20.1	13.1	11.1	4.4
LLM-Grounder (Yang et al., 2024)	×	17.1	5.3	—	—	—	—
ZSVG3D (Yuan et al., 2024b)	×	36.4	32.7	63.8	58.4	27.7	24.6
SeeGround (Li et al., 2025)	×	44.1	39.4	75.7	68.9	34.0	30.0
VLM-Grounder (Xu et al., 2025)	×	51.6	32.8	66.0	29.8	48.3	33.5
OriGround (ours)	×	52.4	42.8	78.3	69.6	37.3	27.2

Table 3: Ablation on the full Nr3D split. Rows progressively add MIP, PGC, and Ori.-Hint to Symbolic Filtering. Overall accuracy (%) is reported.

#	VLM	MIP	PGC	Ori.-Hint	Overall
(a)	×	×	×	×	50.0
(b)	✓	✓	×	×	58.6
(c)	✓	✓	✓	×	60.5
(d)	✓	✓	✓	✓	61.3

Table 4: Perspective-alignment study on the Observer-Oriented subset of Nr3D (Nr3D-VP). The module columns follow the same notation as Table 3.

Method	Modules				Observer-Oriented
	VLM	MIP	PGC	Ori.-Hint	Overall
LaSP	×	×	×	×	45.3
LaSP	✓	×	×	×	47.9
OriGround	✓	✓	×	×	50.9
OriGround	✓	✓	✓	✓	54.7

pruning, which enables VLM to access fine-grained visual features.

- **(c) Perspective-Aligned Prompting.** Our proposed PGC module improves accuracy from 58.6% to 60.5%. This indicates that a viewpoint-aligned top-down map provides useful global context, effectively complementing the limited scope of local candidate images.
- **(d) Orientation-Hint.** Orientation annotations on the aligned map further improve accuracy, confirming that explicit orientation cues help resolve directional relations and spatial understanding.

Effect of Perspective-Alignment. Table 4 examines grounding performance on Nr3D-VP. Starting from Symbolic Filtering, LaSP (Mi et al., 2025) reaches 45.29%. With its own VLM stage, LaSP (Mi et al., 2025) reaches 47.87%. Under the OriGround architecture, enabling MIP module raises accuracy to 50.91%, showing that our candidate construction already yields stronger visual evidence on this subset. Adding PGC module together with Orientation-Hint further lifts accuracy to 54.73%. The gain indicates that aligning the top-down map to the query-consistent observer viewpoint is especially important when the utter-



Figure 5: Qualitative examples from the Observer-Oriented subset of Nr3D (Nr3D-VP). Blue denotes object categories, yellow denotes relation terms, and magenta text highlights observer-oriented expressions.

Table 5: VLM Model ablation on a shared 1000-sample Nr3D probe split.

Method	Model	Overall	Easy	Hard	View Dep.	View Indep.
LaSP	Qwen-VL-Max	50.3	56.2	44.5	45.6	52.4
OriGround	GPT-4o	53.8	62.2	45.7	43.3	58.8
OriGround	Qwen3-VL-Flash	57.0	67.2	46.9	50.0	60.2
OriGround	Qwen-VL-Max	62.3	71.0	53.8	53.4	66.4

ance explicitly specifies the observer’s direction.

Ablation of Vision Language Model. Table 5 separates the effects of architecture and backbone on a 1000-sample probe split randomly sampled from Nr3D to ensure evaluation fairness and representativeness. For a fair architecture comparison, we rerun LaSP (Mi et al., 2025) with Qwen-VL-Max (Bai et al., 2023); under the same VLM model, replacing LaSP with OriGround raises the overall accuracy from 50.3% to 62.3%, showing our proposed orientation-aware symbolic pipeline and visual prompting contribute a substantial gain. Keeping the OriGround architecture fixed and then changing the final VLM further affects performance, from 53.8% with GPT-4o (OpenAI, 2024) to 57.0% with Qwen3-VL-Flash (Bai et al., 2025), with the best result obtained by Qwen-VL-Max at 62.3% overall. Together, these results indicate that OriGround performs best when a stronger VLM is paired with the proposed architecture, allowing the structured visual prompt to be exploited more fully.

4.4 Qualitative Results

Figure 5 provides qualitative evidence for the gains reported in Table 4. Blue text denotes object categories, yellow text denotes relation terms, and magenta text marks observer-oriented expressions

such as *facing the door* or *with your back to the door*. These viewpoint cues define the observer frame under which directional relations should be interpreted.

From the perspective of module design, these examples show why MIP alone is often insufficient on Nr3D-VP: after symbolic filtering, multiple candidates can still remain locally plausible. PGC module addresses this ambiguity by rotating the top-down map into the query-consistent observer frame, while Ori.-Hint further renders object front directions on the aligned map. Together, these two components let the VLM interpret directional relations under the correct perspective, which qualitatively explains the improvement from the system only with MIP module to the full OriGround system.

5 Conclusion

We presented OriGround, an orientation-aware neuro-symbolic framework for zero-shot 3D visual grounding. By parsing language into a viewpoint-aware symbolic program, extracting object-centric orientations, executing orientation-aware geometric reasoning, and aligning the final visual prompt with the inferred observer frame, OriGround resolves view-dependent spatial expressions more reliably than prior training-free approaches. Experiments on Nr3D and ScanRefer show consistent gains over strong training-free baselines, with the largest improvements appearing on observer-oriented queries. These results suggest that explicit orientation modeling is a practical step toward more robust zero-shot grounding in embodied 3D environments.

6 Limitations

Inference Cost. OriGround combines symbolic parsing, orientation estimation, geometric execution, and a final VLM decision stage. This richer pipeline is still less lightweight than simpler training-free alternatives, and the added symbolic processing plus viewpoint-aligned prompting increase both latency and token usage. More details are provided in Appendix C.

Dependence on Upstream Models. OriGround benefits from reliable candidate generation, language parsing, orientation estimation, and final VLM verification. When these upstream components are noisy, errors can propagate through the pipeline and make the final ranking less stable, especially for fine-grained directional distinctions.

Viewpoint Abstraction. Our perspective-aligned top-down map is most natural for horizontal observer-centric relations. It is less expressive for weakly specified viewpoints, vertically structured cases, or spatial cues that require richer 3D context than a floor-plane abstraction can capture.

Symbolic Expressiveness. Like LaSP (Mi et al., 2025), OriGround maps open-ended language into a predefined symbolic vocabulary. This keeps reasoning controllable and efficient, but it can be less flexible for rare or highly compositional expressions that do not map cleanly to the supported predicates.

References

Panos Achlioptas, Ahmed Abdelreheem, Fei Xia, Mohamed Elhoseiny, and Leonidas J. Guibas. 2020. ReferIt3D: Neural listeners for fine-grained 3D object identification in real-world scenes. In *Computer Vision – ECCV 2020*, Lecture Notes in Computer Science, pages 422–440. Springer.

Jacob Andreas, Marcus Rohrbach, Trevor Darrell, and Dan Klein. 2016. Neural module networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 39–48.

Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. 2023. Qwen-VL: A versatile vision-language model for understanding, localization, text reading, and beyond. *arXiv preprint arXiv:2308.12966*.

Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, Wenbin Ge, Zhi-fang Guo, Qidong Huang, Jie Huang, Fei Huang, Binyuan Hui, Shutong Jiang, Zhaohai Li, Mingsheng

Li, and 45 others. 2025. Qwen3-VL technical report. *arXiv preprint arXiv:2511.21631*.

Eslam Mohamed Bakr, Mohamed Ayman, Mahmoud Ahmed, Habib Slim, and Mohamed Elhoseiny. 2024. CoT3DRef: Chain-of-thoughts data-efficient 3D visual grounding. In *The Twelfth International Conference on Learning Representations*.

Dave Zhenyu Chen, Angel X. Chang, and Matthias Nießner. 2020. ScanRefer: 3D object localization in RGB-D scans using natural language. In *Computer Vision – ECCV 2020*, Lecture Notes in Computer Science, pages 202–221. Springer.

Shizhe Chen, Pierre-Louis Guhur, Makarand Tapaswi, Cordelia Schmid, and Ivan Laptev. 2022. Language conditioned spatial relation reasoning for 3D object grounding. In *Advances in Neural Information Processing Systems*.

Jiading Fang, Xiangshan Tan, Shengjie Lin, Igor Vasiljevic, Vitor Guizilini, Hongyuan Mei, Rares Ambrus, Gregory Shakhnarovich, and Matthew R. Walter. 2024. Transcrib3D: 3D referring expression resolution through large language models. *Preprint, arXiv:2404.19221*.

Chun Feng, Joy Hsu, Weiyu Liu, and Jiajun Wu. 2024. Naturally supervised 3D visual grounding with language-regularized concept learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13269–13278.

Joy Hsu, Jiayuan Mao, and Jiajun Wu. 2023. NS3D: Neuro-symbolic grounding of 3D objects and relations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2614–2623.

Pin-Hao Huang, Han-Hung Lee, Hwann-Tzong Chen, and Tyng-Luh Liu. 2021. Text-guided graph neural networks for referring 3D instance segmentation. In *Proceedings of the AAAI Conference on Artificial Intelligence*.

Ayush Jain, Nikolaos Gkanatsios, Ishita Mediratta, and Katerina Fragkiadaki. 2022. Bottom up top down detection transformers for language grounding in images and point clouds. In *Computer Vision – ECCV 2022*, Lecture Notes in Computer Science, pages 417–433. Springer.

Zhao Jin, Rong-Cheng Tu, Jingyi Liao, Wenhao Sun, Xiao Luo, Shunyu Liu, and Dacheng Tao. 2025. SPAZER: Spatial-semantic progressive reasoning agent for zero-shot 3D visual grounding. In *Advances in Neural Information Processing Systems*.

Rong Li, Shijie Li, Lingdong Kong, Xulei Yang, and Junwei Liang. 2025. SeeGround: See and ground for zero-shot open-vocabulary 3D visual grounding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3707–3717.

643	Jiayuan Mao, Chuang Gan, Pushmeet Kohli, Joshua B	<i>Conference on Computer Vision and Pattern Recog-</i>	699
644	Tenenbaum, and Jiajun Wu. 2019. The neuro-	nition, pages 20623–20633.	700
645	symbolic concept learner: Interpreting scenes, words,		
646	and sentences from natural supervision. <i>arXiv</i>		
647	<i>preprint arXiv:1904.12584</i> .		
648	Boyun Mi, Hanqing Wang, Tai Wang, Yilun Chen, and	Zhihao Yuan, Xu Yan, Yinghong Liao, Yao Guo, Guan-	701
649	Jiangmiao Pang. 2025. Language-to-space program-	bin Li, Shuguang Cui, and Zhen Li. 2021. In-	702
650	ming for training-free 3D visual grounding . In <i>Pro-</i>	stanceRefer: Cooperative holistic understanding for	703
651	<i>ceedings of the 2025 Conference on Empirical Meth-</i>	visual grounding on point clouds through instance	704
652	<i>ods in Natural Language Processing</i> , pages 3844–	multi-level contextual referring. In <i>Proceedings of</i>	705
653	3864, Suzhou, China. Association for Computational	<i>the IEEE/CVF International Conference on Com-</i>	706
654	Linguistics.	<i>puter Vision</i> .	707
655	OpenAI. 2024. GPT-4o System Card . Accessed: 2026-	Lichen Zhao, Daigang Cai, Lu Sheng, and Dong Xu.	708
656	05-19.	2021. 3DVG-Transformer: Relation modeling for vi-	709
		sual grounding on point clouds. In <i>Proceedings of the</i>	710
657	Songyou Peng, Kyle Genova, Chiyu Jiang, An-	<i>IEEE/CVF International Conference on Computer</i>	711
658	drea Tagliasacchi, Marc Pollefeys, and Thomas	<i>Vision</i> .	712
659	Funkhouser. 2023. OpenScene: 3D scene under-		
660	standing with open vocabularies. In <i>Proceedings of</i>	Ziyu Zhu, Xiaojian Ma, Yixin Chen, Zhidong Deng,	713
661	<i>the IEEE/CVF Conference on Computer Vision and</i>	Siyuan Huang, and Qing Li. 2023. 3D-VisTA: Pre-	714
662	<i>Pattern Recognition</i> .	trained transformer for 3D vision and text alignment.	715
		In <i>Proceedings of the IEEE/CVF International Con-</i>	716
663	Zehan Wang, Ziang Zhang, Jiayang Xu, Jialei Wang,	<i>ference on Computer Vision</i> , pages 2911–2921.	717
664	Tianyu Pang, Chao Du, Hengshuang Zhao, and Zhou		
665	Zhao. 2025. Orient Anything V2: Unifying orienta-	This appendix provides supplementary details	718
666	tion and rotation understanding. In <i>The Thirty-ninth</i>	for OriGround: Section A presents qualitative rea-	719
667	<i>Annual Conference on Neural Information Process-</i>	soning cases, Section B details orientation extrac-	720
668	<i>ing Systems</i> .	tion and robustness, Section C reports token usage,	721
		and Section D lists the main prompt templates.	722
669	Runsen Xu, Zhiwei Huang, Tai Wang, Yilun Chen,	A Representative Reasoning Cases	723
670	Jiangmiao Pang, and Dahua Lin. 2025. VLM-		
671	Grounder: A VLM agent for zero-shot 3D visual	We present four case studies that expose the raw	724
672	grounding . In <i>Proceedings of the 8th Conference on</i>	symbolic output, executor scores, visual prompts,	725
673	<i>Robot Learning</i> , volume 270 of <i>Proceedings of Ma-</i>	and final VLM reasoning evidence.	726
674	<i>chine Learning Research</i> , pages 3961–3985. PMLR.		
675	Jianing Yang, Xuweiyi Chen, Shengyi Qian, Nikhil	A.1 Observer-Oriented Example 1	727
676	Madaan, Madhavan Iyengar, David F. Fouhey, and	Original Instruction	728
677	Joyce Chai. 2024. LLM-Grounder: Open-vocabulary	{	729
678	3D visual grounding with large language model as an	"target_id": 3,	730
679	agent. In <i>IEEE International Conference on Robotics</i>	"sentence": "If you face the whiteboard, the	731
680	<i>and Automation (ICRA)</i> .	chair is the one on the left closer to the	732
		board."	733
681	Zhengyuan Yang, Songyang Zhang, Liwei Wang, and	}	734
682	Jiebo Luo. 2021. SAT: 2D semantics assisted train-	Symbolic Parser Output	735
683	ing for 3D visual grounding. In <i>Proceedings of the</i>	{	736
684	<i>IEEE/CVF International Conference on Computer</i>	"json_obj": {	737
685	<i>Vision</i> , pages 1856–1866.	"category": "chair",	738
		"relations": [	739
686	Kexin Yi, Jiajun Wu, Chuang Gan, Antonio Torralba,	{	740
687	Pushmeet Kohli, and Josh Tenenbaum. 2018. Neural-	"relation_name": "left",	741
688	symbolic VQA: Disentangling reasoning from vision	"objects": [	742
689	and language understanding. In <i>Advances in Neural</i>	{	743
690	<i>Information Processing Systems</i> .	"category": "whiteboard",	744
		"relations": []	745
691	Qihao Yuan, Jiaming Zhang, Kailai Li, and Rainer	}]	746
692	Stiefelhagen. 2024a. Solving zero-shot 3D visual	}]	747
693	grounding as constraint satisfaction problems. <i>arXiv</i>	},	748
694	<i>preprint arXiv:2411.14594</i> .	{	749
695	Zhihao Yuan, Jinke Ren, Chun-Mei Feng, Hengshuang	"relation_name": "closer",	750
696	Zhao, Shuguang Cui, and Zhen Li. 2024b. Visual	"objects": [	751
697	programming for zero-shot open-vocabulary 3D vi-	{	752
698	sual grounding. In <i>Proceedings of the IEEE/CVF</i>	"category": "whiteboard",	753

```

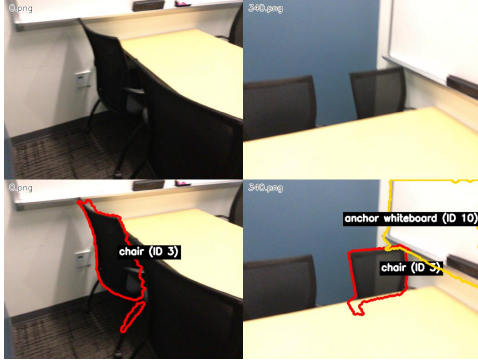
    "relations": []
  }
}
],
},
"is_viewpoint_instruction": true,
"viewpoint_anchor_phrases": ["whiteboard"],
"viewpoint_anchor_categories": ["whiteboard"],
"viewpoint_anchor_front_relation": "
  opposite_front"
}

```

Executor Top-5 Selection

Rank	1	2	3	4	5
ID	3	1	11	2	4
Cat.	chair	chair	chair	chair	table
Score	1.000	0.870	0.540	0.486	2.23e-6

Visual Prompt



Instance-Centric Prompting. The candidate view centers on the correct target object.



Perspective-Aligned Prompting. Top-down map under the predicted observer viewpoint.

VLM Reasoning

```

{
  "reasoning": "The target description specifies
    facing the whiteboard and identifying the
    chair on the left closer to it. The top-down
    map shows the observer viewpoint (green
    arrow) facing the whiteboard (anchor 10).
    Chair ID 3 is positioned to the left of the
    whiteboard from this viewpoint and is closer
    to it than other chairs. The candidate
    image for ID 3 confirms its proximity to the
    whiteboard and its position on the left
    side when facing the board. Other chairs (ID
    1, 2, 11) are either farther or not on the
    left side relative to the whiteboard.",
  "object id": 3
}

```

A.2 Observer-Oriented Example 2

Original Instruction

```

{
  "target_id": 20,
  "sentence": "The lamp to the right if you are
    lying on the bed. The lamp closest to the
    doll house."
}

```

Symbolic Parser Output

```

{
  "json_obj": {
    "category": "lamp",
    "relations": [
      {
        "relation_name": "right",
        "objects": [
          {
            "category": "bed",
            "relations": []
          }
        ]
      },
      {
        "relation_name": "closest",
        "objects": [
          {
            "category": "doll house",
            "relations": []
          }
        ]
      }
    ]
  },
  "is_viewpoint_instruction": true,
  "viewpoint_anchor_phrases": ["bed"],
  "viewpoint_anchor_categories": ["bed"],
  "viewpoint_anchor_front_relation": "same_front"
}

```

Executor Top-5 Selection

Rank	1	2	3	4	5
ID	19	20	1	0	2
Cat.	lamp	lamp	book	books	trash can
Score	1.000	0.480	5.37e-6	1.23e-6	1.86e-7

823

Visual Prompt



Instance-Centric Prompting. The candidate view centers on the correct target object.



Perspective-Aligned Prompting. Top-down map under the predicted observer viewpoint.

824

VLM Reasoning

```
{
  "reasoning": "The target description specifies
    the lamp to the right when lying on the bed
    , and the one closest to the doll house.
    From the top-down map, the observer
    viewpoint (green arrow) is facing the bed,
    with the bed's front direction indicated by
    the yellow arrow. Lamp ID 20 is positioned
    to the right of the bed from this viewpoint
    and is adjacent to the doll house, as seen
    in the candidate image for ID 20. Lamp ID 19
    is on the left side of the bed. The doll
    house is clearly visible next to lamp ID 20
    in the annotated views, confirming proximity
    . Thus, lamp ID 20 satisfies both spatial
    conditions.",
  "object id": 20
}
```

843

A.3 Observer-Oriented Example 3

844

Original Instruction

{ 845

"target_id": 12, 846

"sentence": "When standing at the foot of the 847

bed, the pillow in the middle on the left 848

hand side" 849

} 850

Symbolic Parser Output

851

{ 852

"json_obj": { 853

"category": "pillow", 854

"relations": [855

{ 856

"relation_name": "middle", 857

"objects": [858

{ 859

"category": "bed", 860

"relations": [] 861

} 862

] 863

}, 864

{ 865

"relation_name": "left", 866

"objects": [867

{ 868

"category": "bed", 869

"relations": [] 870

} 871

] 872

} 873

] 874

}, 875

"is_viewpoint_instruction": true, 876

"viewpoint_anchor_phrases": ["the foot of the 877

bed"], 878

"viewpoint_anchor_categories": ["bed"], 879

"viewpoint_anchor_front_relation": " 880

opposite_front" 881

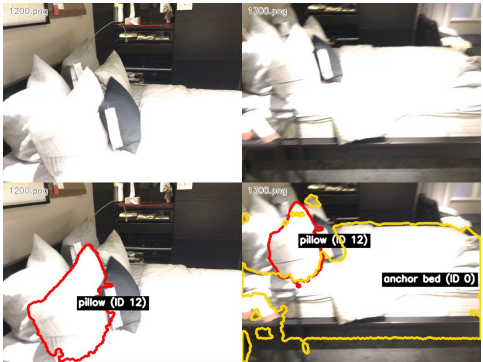
} 882

Executor Top-5 Selection

Rank	1	2	3	4	5
ID	12	15	11	13	14
Cat.	pillow	pillow	pillow	pillow	pillow
Score	1.000	0.922	0.788	0.588	0.588

Visual Prompt

883



Instance-Centric Prompting. The candidate view centers on the correct target object.



Perspective-Aligned Prompting. Top-down map under the predicted observer viewpoint.

VLM Reasoning

```
{
  "reasoning": "The target description specifies
    standing at the foot of the bed and
    identifying the pillow in the middle on the
    left-hand side. The top-down map shows the
    observer viewpoint (green arrow) facing the
    head of the bed, with the foot of the bed at
    the bottom of the map. From this
    perspective, the left side of the bed
    corresponds to the left side of the map.
    Among the pillows, ID 12 is positioned
    centrally along the left side of the bed
    when viewed from the foot. The candidate
    image for ID 12 shows it located in the
    middle-left cluster of pillows, consistent
    with the description. Other candidates like
    ID 15 are more toward the far left, and IDs
    13 and 14 are on the right side, making ID
    12 the best match.",
  "object id": 12
}
```

```

    {
      "relation_name": "above",
      "objects": [
        {
          "category": "tv",
          "relations": []
        }
      ],
      "negative": true
    },
    {
      "relation_name": "close",
      "objects": [
        {
          "category": "opening",
          "relations": []
        }
      ]
    },
    {
      "relation_name": "close",
      "objects": [
        {
          "category": "bed",
          "relations": []
        }
      ]
    }
  ]
},
"is_viewpoint_instruction": false,
"viewpoint_anchor_phrases": [],
"viewpoint_anchor_categories": [],
"viewpoint_anchor_front_relation": "unknown"
}

```

Executor Top-5 Selection

Rank	1	2	3	4	5
ID	0	3	14	1	76
Cat.	backpack	backpack	backpack	bag	toilet paper
Score	1.000	0.951	0.937	7.27e-4	8.86e-7

A.4 Nested Spatial Relation Example

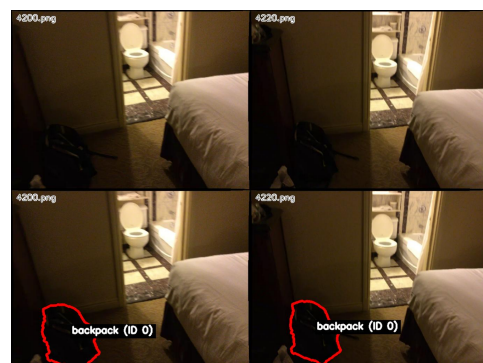
Original Instruction

```
{
  "target_id": 0,
  "sentence": "black backpack to right of the
               dresser with tv on it nearest the opening to
               the bathroom and the end of the bed,"
}
```

Symbolic Parser Output

```
{
  "json_obj": {
    "category": "backpack",
    "relations": [
      {
        "relation_name": "right",
        "objects": [
          {
            "category": "dresser",
            "relations": [
```

Visual Prompt



Instance-Centric Prompting. The candidate view centers on the correct target object.

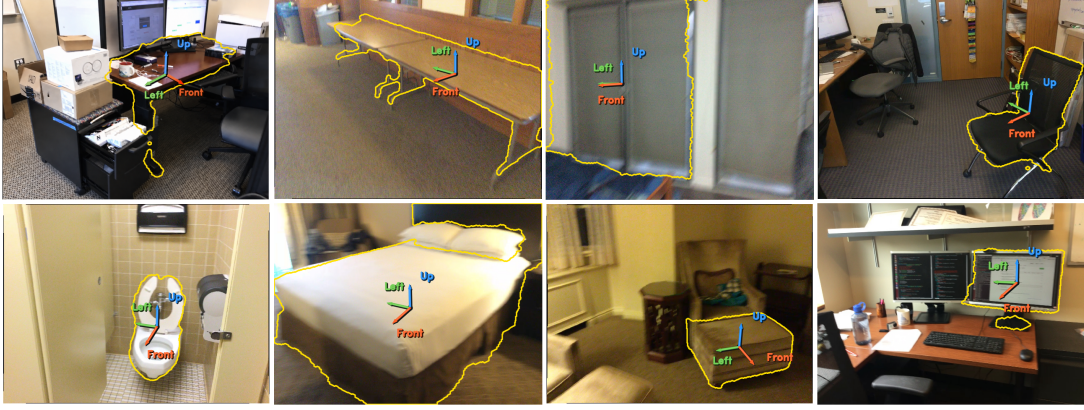
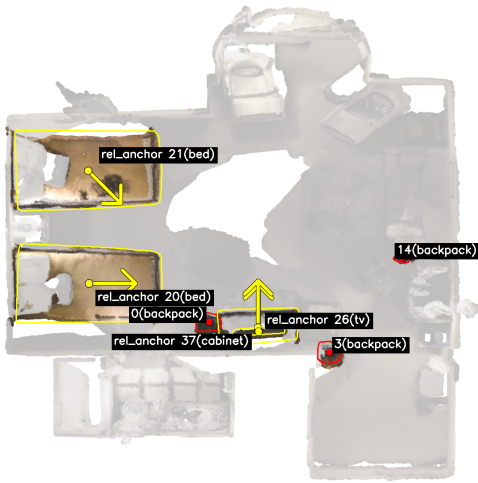


Figure 6: Object-centric orientation extraction examples produced by Orient-Anything-V2.



Perspective-Aligned Prompting. The original query does not contain an explicit observer viewpoint, so the top-down map provides only global layout information.

VLM Reasoning

```
{
  "reasoning": "The target description specifies a black backpack to the right of a dresser with a TV on it, near the bathroom opening and the end of the bed. In the top-down map, ID 0 is positioned near the bed and bathroom entrance, adjacent to the cabinet (ID 37) which has a TV (ID 26) on it. The annotated images for ID 0 show a black backpack located to the right of the dresser with the TV, consistent with the description. The other candidates (IDs 3, 14, 1, 76) are either not near the specified anchor objects or do not match the spatial relationship described. ID 76 is toilet paper, not a backpack. Thus, ID 0 best matches the target description.",
  "object id": 0
}
```

B Experimental Details

Basic evaluation settings. We keep the top-5 candidates after symbolic execution and set the filtering threshold to 0.0. In the VLM stage, the decoding hyper-parameters are temperature = 0.1 and top- p = 0.2. For the evaluation on ReferIt3D, we use the object classification results from ZSVG3D (Yuan et al., 2024b).

Orientation extraction details. Figure 6 visualizes the object-centric front directions estimated by Orient-Anything-V2 for several scene objects. Each masked object crop is passed to the orientation model, and the predicted front direction is then projected back to scene coordinates before being used by symbolic scoring and viewpoint-aligned prompting.

Robustness to orientation noise. Although the estimated object orientation can be imperfect, the resulting error is often not catastrophic in our pipeline. In many cases, if the target category is correct, the orientation-aware symbolic executor still assigns the target a non-trivial score and keeps it within the top-5 candidate set. This preserves recall for the final VLM stage, which can then use instance appearance together with the aligned global context to verify whether the candidate truly satisfies the query. In this sense, the orientation module acts less like a brittle hard constraint and more like a useful ranking signal that improves robustness even when its estimates are not exact.

Below we show two recovered examples where the correct target is retained in the top-5 list but ranked last by the symbolic executor. The final VLM stage still resolves the target correctly by combining the candidate-centric views with the aligned scene-level evidence.

B.1 Robustness Recovery Example 1

Original Instruction

```
{
  "target_id": 66,
  "sentence": "Facing the shower stall, it's the
               bottle on the left."
}
```

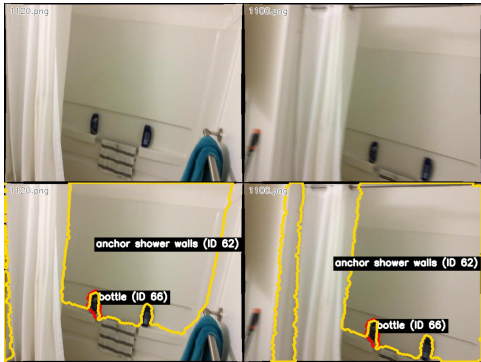
Symbolic Parser Output

```
{
  "json_obj": {
    "category": "bottle",
    "relations": [
      {
        "relation_name": "left",
        "objects": [
          {
            "category": "shower stall",
            "relations": []
          }
        ]
      }
    ]
  },
  "is_viewpoint_instruction": true,
  "viewpoint_anchor_phrases": ["shower stall"],
  "viewpoint_anchor_categories": ["shower stall"],
  "viewpoint_anchor_front_relation": "
    opposite_front"
}
```

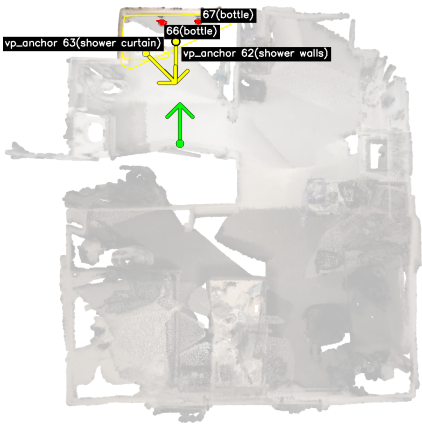
Executor Top-5 Selection

Rank	1	2	3	4	5
ID	67	35	22	31	66
Cat.	bottle	jacket	ceiling fan	shoes	bottle
Score	1.000	3.20e-6	1.30e-6	9.76e-7	4.15e-7

Visual Prompt



Instance-Centric Prompting. The candidate view centers on the correct target object.



Perspective-Aligned Prompting. Top-down map under the predicted observer viewpoint.

VLM Reasoning

```
{
  "reasoning": "The target description specifies
                'facing the shower stall, it's the bottle
                on the left.' The top-down map shows two
                bottles (ID 66 and ID 67) near the shower
                walls (anchor ID 62). The green arrow
                indicates the observer is facing the shower
                stall. In the candidate images, ID 66 is
                positioned to the left of the shower stall
                when facing it, while ID 67 is on the right.
                The annotated views confirm that ID 66 is
                the left bottle relative to the shower stall
                from the observer's perspective. Therefore,
                ID 66 matches the description.",
  "object id": 66
}
```

B.2 Robustness Recovery Example 2

Original Instruction

```
{
  "target_id": 18,
  "sentence": "it is the bottle on the right-
               hand side of the sink"
}
```

Symbolic Parser Output

```
{
  "json_obj": {
    "category": "bottle",
    "relations": [
      {
        "relation_name": "right",
        "objects": [
          {
            "category": "sink",
            "relations": []
          }
        ]
      }
    ]
  },
  "is_viewpoint_instruction": true,
  "viewpoint_anchor_phrases": ["sink"],
  "viewpoint_anchor_categories": ["sink"],
}
```



```

1095 "viewpoint_anchor_front_relation": "
1096     opposite_front"
1097 }

```

Executor Top-5 Selection

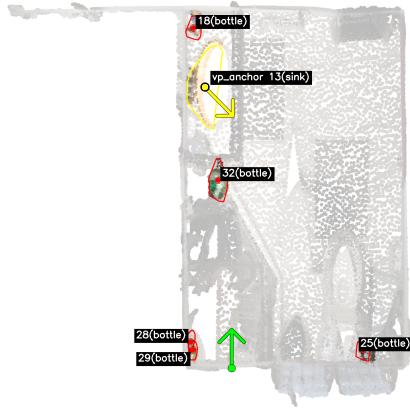
Rank	1	2	3	4	5
ID	25	32	28	29	18
Cat.	bottle	bottle	bottle	bottle	bottle
Score	1.000	0.905	0.905	0.905	3.54e-5

1098

Visual Prompt



Instance-Centric Prompting. The candidate view centers on the correct target object.



Perspective-Aligned Prompting. Top-down map under the predicted observer viewpoint.

1099

VLM Reasoning

```

1100 {
1101   "reasoning": "The target description specifies
1102     'the bottle on the right-hand side of the
1103     sink'. The top-down map shows the sink (ID
1104     13) with bottle ID 18 positioned to its
1105     right. In the candidate image for ID 18, the
1106     bottle is clearly visible on the right side
1107     of the sink from the observer's viewpoint,
1108     consistent with the green arrow indicating
1109     the observer's direction. Other bottles (IDs

```

```

25, 28, 29, 32) are not located near the
sink or on its right side. Thus, ID 18 best
matches the description.",
"object id": 18
}

```

1110
1111
1112
1113
1114

C Token Usage Analysis

Table 6 summarizes the per-sample token usage comparison following the presentation style of LaSP (Mi et al., 2025). We additionally include OriGround based on our token accounting run.

1115
1116
1117
1118
1119

Method	Time/s	Token
ZSVG3D (Yuan et al., 2024b)	2.4	2.5k
VLM-Grounder (Xu et al., 2025)	50.3	8k
Transcrib3D (Fang et al., 2024)	27.0	12k
CSVG (Yuan et al., 2024a)	4.0	4.0k
SeeGround (Li et al., 2025)	9.0	2.6k
LaSP (Mi et al., 2025) (w/o VLM)	2.1	1.2k
LaSP (Mi et al., 2025)	7.7	3.1k
OriGround	10.2	7.1k

Table 6: Per-sample token usage comparison.

For OriGround, symbolic processing consumes about 2.58k tokens per sample, corresponding to 2.43M prompt tokens and 0.14M completion tokens in total. The downstream visual reasoning stage consumes about 4.48k tokens per sample, corresponding to 4.34M prompt tokens and 0.13M completion tokens in total. Summing these two stages yields about 7.06k tokens per sample, which we report as 7.1k in Table 6. The reported time of 10.2 seconds is the end-to-end total, including 4.2 seconds for symbolic processing and 6.0 seconds for the subsequent visual reasoning stage.

1120
1121
1122
1123
1124
1125
1126
1127
1128
1129
1130
1131

D Prompts

In this section, we show the key prompts used by OriGround. We organize them into two parts: symbolic prompts and reasoning prompts. Listings 1 and 2 present the two symbolic stages, namely relation decomposition and observer-viewpoint inference. Listing 3 presents the reasoning prompt for viewpoint-aligned visual grounding.

Prompt for semantic parsing.

```
You are a parser for 3D referring expressions.

## Objective
Convert a natural-language query into a normalized relation tree with a target object category
and a set of canonical spatial predicates.

## Output Schema
Return STRICT JSON only:
{
  "category": "target category",
  "relations": [...]
}

Each relation entry must use:
- "relation_name": canonical predicate
- "objects": referenced object list
- "negative": optional boolean

## Predicate Inventory
Use only the following relation names.

Unary:
["corner", "on the floor", "against wall", "smaller", "larger", "taller", "lower", "within"]

Binary:
["above", "below", "beside", "close", "far", "left", "right", "front", "behind", "across"]

Ternary:
["between", "center", "middle"]

## Parsing Rules
1. The root "category" is the target object category named by the query.
2. Normalize surface wording to the closest canonical predicate above.
3. Never invent a new relation name.
4. Keep predicates simple and reusable. Do not create relation strings such as "left of a blue
   box" or "facing the window".
5. Encode nested descriptions recursively inside the referenced objects.
6. For unary relations, use "objects": [].
7. For binary relations, provide exactly one referenced object.
8. For ternary relations, provide exactly two referenced objects.
9. If the sentence explicitly negates a relation, set "negative": true for that relation.
10. Ignore observer-viewpoint phrases here; they are handled in the next symbolic stage.

## Recommended Procedure
- Identify the target category.
- Extract the minimal set of spatial relations needed to distinguish the target.
- Build referenced objects recursively.
- Reason internally if needed, but output JSON only.

## Example
Sentence: The correct whiteboard is the one on a table.

Output:
```json
{
 "category": "whiteboard",
 "relations": [
 {
 "relation_name": "above",
```

```

 "objects": [
 {
 "category": "table",
 "relations": []
 }
]
 }
]
}
...

```

3 more examples...

## Prompt for observer-viewpoint inference.

You are doing stage 2 of NR3D parsing: observer-viewpoint inference only.

### ## Goal

Decide whether the sentence defines an explicit or implicit observer frame, and if so, extract the viewpoint anchor and its front-direction relation.

### ## Input

- sentence: {sentence}
- stage1\_json\_obj: {json\_obj\_text}

### ## Output

Return STRICT JSON only with the schema:

```

{
 "is_viewpoint_instruction": true,
 "viewpoint_extraction_reason": "short reason",
 "viewpoint_anchor_phrases": ["phrase1"],
 "viewpoint_anchor_categories": ["category1"],
 "viewpoint_anchor_front_relation": "same_front"
}

```

### ## Core Constraints

1. Use the sentence as the primary evidence.
2. Use `stage1\_json\_obj` only as frozen context for the target category and relation anchors. Do not rewrite the stage-1 symbolic parse.
3. Preserve the stage-1 target category. Do not switch the target to an anchor category.
4. Keep `viewpoint\_extraction\_reason` short.
5. Anchor phrases should be copied from the sentence when possible.
6. Anchor categories should be normalized object categories usable downstream.

### ## Positive Cases

Set `is\_viewpoint\_instruction=true` for explicit observer-frame expressions such as:

- `facing ...`
- `looking at ...`
- `looking toward ...`
- `head on`
- `from the foot of the bed`
- `if you are sitting on ...`

Also mark `true` for implicit group-ordering cases when the sentence resolves one object in a group using an ordering or directional frame, for example:

- `the middle box of the stack`
- `the top one`
- `the blanket above the other blanket`
- `the narrower of the two stalls`
- `the back pillow on the right`
- `the last on the right`

When no explicit external anchor exists in such group-ordering cases, it is acceptable to use the target category itself as `viewpoint\_anchor\_categories`.

### ## Front-Relation Policy

Valid values of `viewpoint\_anchor\_front\_relation` are only: `same\_front`, `opposite\_front`, `left\_of\_front`, `right\_of\_front`, `unknown`

Typical `same\_front` cases:

- `sitting on ...`
- `sitting in ...`
- `through the doorway`
- some `standing at ...` cases where the observer shares the anchor's front direction

Typical `opposite\_front` cases:

- `facing ...`
- `looking at ...`
- `head on`
- many group-ordering cases without an explicit external anchor

**## Negative Cases**

Pure object-object relations usually stay `false` when they do not define an observer frame or a group-ordering frame, for example:

- `farthest from the door`
- `next to the desk`
- `close to the window`

When `is\_viewpoint\_instruction=false`, output:

- `viewpoint\_anchor\_phrases: []`
- `viewpoint\_anchor\_categories: []`
- `viewpoint\_anchor\_front\_relation: "unknown"`

**## Examples**

Example 1

sentence: Facing the desk, the book on the left.

stage1\_json\_obj:

```
{
 "category": "book",
 "relations": [
 {
 "relation_name": "left",
 "objects": [
 {
 "category": "desk",
 "relations": []
 }
]
 }
]
}
```

output:

```
{
 "is_viewpoint_instruction": true,
 "viewpoint_extraction_reason": "Facing the desk defines a fixed observer direction.",
 "viewpoint_anchor_phrases": ["the desk"],
 "viewpoint_anchor_categories": ["desk"],
 "viewpoint_anchor_front_relation": "opposite_front"
}
```

3 more examples...

1142

## Prompt for viewpoint-aligned visual grounding.

1143

You are a visual grounding expert for indoor scenes.

### ## Input

You will receive multiple images.

### ## Image Order

- The first image is a TOP-DOWN annotation-only map: plain background with vector overlays(availability: {has\_topdown\_map}, type: {topdown\_map\_kind}).
- The remaining images are one per candidate object.

### ## Sample Type

For this sample type, the instruction contains a fixed observer viewpoint cue.

### ## Top-Down Map Semantics

Top-down is annotation-only (without\_pcd). The semantics are:

- GREEN arrow(s) encode hypothesized observer viewing direction.
- YELLOW arrow near reference anchor indicates anchor front direction estimate.
- RED / YELLOW regions indicate candidate / anchor boundaries.
- MAGENTA regions indicate overlap objects that are both candidate and anchor.
- Candidates are labeled as `(ID x) category`; anchors are labeled as `anchor (ID x) category`.
- Candidate/anchor centers are marked with red/yellow circles for relative left/right/front/behind comparison.

### ## Candidate Image Layout

1144

Each candidate image has a 2x2 layout:

- Top row: two original full-frame views (raw scene context)
- Bottom row: the same two views with annotations

### ## Candidate Annotation Semantics

Annotation semantics (critical):

- The candidate is highlighted with a red translucent region + red boundary contour and labeled as `category (ID x)`.
- Anchor/reference objects (if visible in the same frame) are highlighted and labeled as `anchor <category> (ID x)`.
- Anchor annotations are relational cues only, not automatic targets.
- Do NOT confuse candidate and anchor: red annotation indicates candidate; yellow annotation indicates anchor-only reference.

### ## Task

Select exactly one object ID from the candidate list that best matches the target description.

### ## Input Fields

Target Object Description: "{utterance}"

Candidate list: {candidates}

### ## Decision Policy

- 0) Candidate vs anchor disambiguation (mandatory)
  - First verify annotation color semantics before reasoning: candidate is red; anchor is non-red reference annotation.
  - Never output an object only because it appears as an anchor marker.
- 1) Scene-image-first grounding
  - Candidate images (raw + annotated views) are primary evidence.
  - Top-down is auxiliary for global horizontal layout and viewpoint-conditioned directional cross-check.
- 2) Fixed-viewpoint alignment from language (critical)
  - Parse the sentence for the viewpoint cue (e.g., facing/looking at/standing at the foot of the bed).
  - Use top-down directional cues to instantiate the observer frame, then evaluate left/right/front/back.
  - If directional cues conflict, prefer the interpretation most consistent with the sentence viewpoint cue and candidate-view evidence.
- 3) Top-down reliability handling
  - Orientation arrows can be noisy; allow small geometric tolerance.
  - If the top-down directional signal strongly conflicts with candidate-view evidence, trust candidate-view evidence.
- 4) Top-down limitation
  - Top-down supports horizontal floor-plane reasoning only.
  - Do not use top-down for vertical relations such as on/under, above/below, or height.
- 5) Final selection
  - Choose exactly one candidate ID with the strongest joint evidence from category, relation, and multi-view consistency.
  - Always output one ID from Candidate list, even under uncertainty.

### ## Output Format

- Return STRICT JSON only (no markdown, no extra text):

```
{
 "reasoning": "brief evidence-focused explanation",
 "object id": 0,
 "confidence": 0.0
}
```

For queries without viewpoint instructions, we use the same global-context reasoning template but do not instantiate a fixed observer frame; the top-down map is then treated only as auxiliary global context for candidate distribution and anchor relations, rather than as a committed directional frame.